Fiscal Year:	FY 2005	**Task Last Updated:**	FY 10/24/2006
PI Name:	Noca, Flavio Ph.D.		
Project Title:	Nanowick Circuits For Space Fluid Applications		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Interfacial phenomena		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	flavio@caltech.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	(626) 395-4747
Organization Name:	California Institute of Technology		
PI Address 1:	Division of Bioengineering, Mail Stop: 205-45		
PI Address 2:	1200 E. California Boulevard		
PI Web Page:			
City:	Pasadena	**State:**	CA
Zip Code:	91125	**Congressional District:**	29
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2002 Physical Sciences 02-OBPR-03-C
Start Date:	10/01/2004	**End Date:**	09/30/2006
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Added to Task Book in 10/2006 per GRC		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Bronikowski, Michael (JPL)		
Grant/Contract No.:	NNC05GA66G		
Performance Goal No.:			
Performance Goal Text:			
	A wick is thread, strip, or bundle of fibers that, by capillary action, draws up the oil of a lamp or the melted wax in a candle to be burned (Encyclopedia Britannica Online). We have fabricated surface patterns of dense arrays of aligned carbon nanotubes and have used them as “highways” for liquids to flow at high speeds (centimeters per second) by wicking action through the interstices of the dense nanotube mats. These nanofiber strips, or nanowicks, remove a long-standing roadblock in the field of micro/nano-fluidics engineering, namely the unreasonably large pressures required to push liquids though small channels. The nanowick not only embodies a revolutionary technological paradigm of novel fluid transport devices running without any power input, moving parts or external pump – which could be especially useful for the extended operation of body-worn or field-installed (bio)fluid sampling devices –, but its intrinsically nanofibrous nature make it also attractive for		

Task Description:	applications beyond simple liquid transport, such as biomolecular ultrafiltration and nanofluidic mixing without any confining channel walls. In the first year, we will focus on an understanding of the fluid mechanics of nanowicking: i. fundamentals of two-dimensional nanoscale wicking, i.e, capillary fluid flow transversely to pre-patterned and well-aligned nanofibers; ii. effects of gravity on nanowicking flows, in order to establish a design envelope for nanowick-based space devices; iii. investigation of various nanowick components along pre-patterned fiber arrays, such as nanoscale mixers and branches. In the second year, we will establish proof-of-concepts of nanowick-based fluidic devices, such as: i. Micro Capillary Pumped Loops (MCPL) for applications such as spacecraft electronic component cooling applications; ii. sampling and delivery fluidic chips on a flexible substrate - “bandwicks” - for body-worn applications; iii. biomolecular ultrafiltration chips.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	New project for FY2005; no report received during this reporting period. New to Task Book in 2006.
Bibliography Type:	Description: (Last Updated:)